

COD - AT 1

M. Vimal

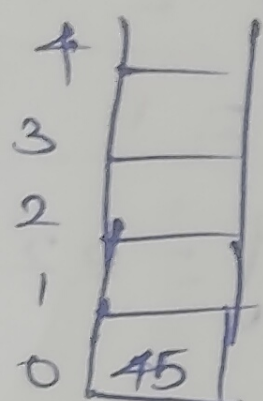
19351944

CSA0303 - Data Structure

30/7/2026

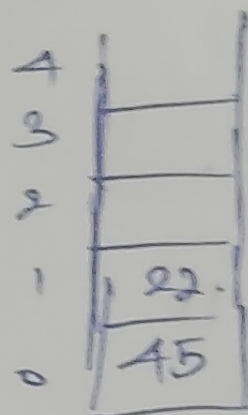
1. Perform the following operations in LIFO data structure having an array size of 5. Push(45), Push(22), Push(11), Push(77), Pop(), Push(44), Push(33). Print the peak element of data structure and its index position.

Ans: i) Push(45).



$$\begin{aligned} \text{push} &= \text{top} + 1 \\ &= -1 + 1 \\ &= 0 \\ \text{Stack} &= [45] \end{aligned}$$

ii) Push(22)



$$\begin{aligned} \text{push} &= \text{top} + 1 \\ &= 0 + 1 \\ &= 1 \\ \text{Stack} &= [22] \end{aligned}$$

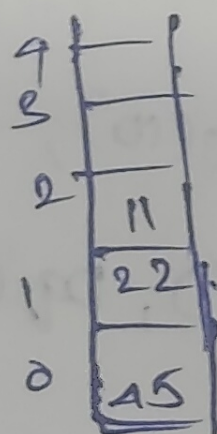
iii) Push

$$= \text{top} + 1$$

$$= 1 + 1$$

$$= 2$$

$$\text{Stack} = [11]$$



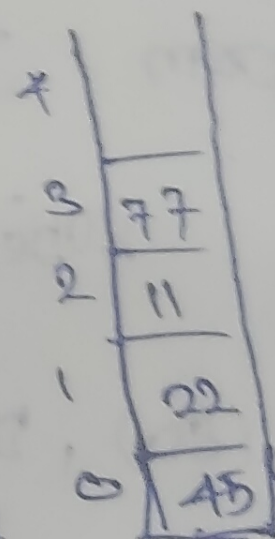
iv) Push(77)

$$= \text{top} + 1$$

$$= 2 + 1$$

$$= 3$$

$$[\text{Stack}] = 77$$



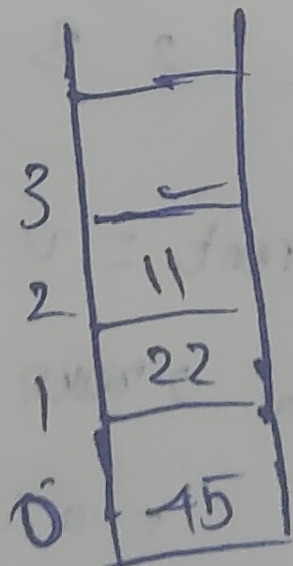
v) Pop()

$$\text{Pop} = \text{top} - 1$$

$$= 3 - 1$$

$$= 2$$

$$\text{Stack} = [11]$$



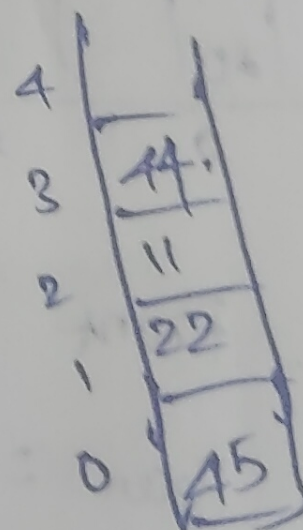
vi) Push(44)

$$= \text{top} + 1$$

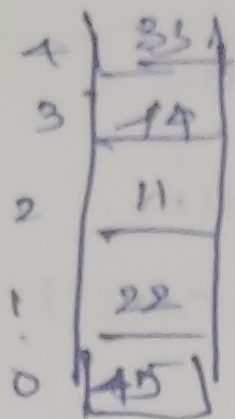
$$= 2 + 1$$

$$= 3$$

$$\text{Stack} = [44]$$



vii) Push (33)



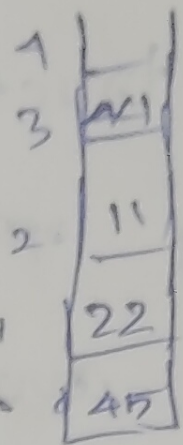
Push = top + 1

$$= 3 + 1$$

$$= 4$$

$$[Stack] = 33$$

viii) Pop()



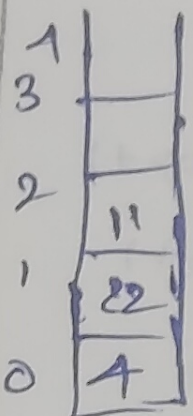
Pop = top - 1

$$= 4 - 1$$

$$= 3$$

$$[Stack] = 22$$

ix) Pop()



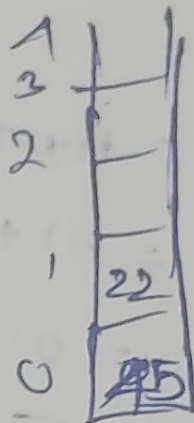
Pop = top - 1

$$= 3 - 1$$

$$= 2$$

$$[Stack] = 11$$

x) Pop()



Pop = top - 1

$$= 2 - 1$$

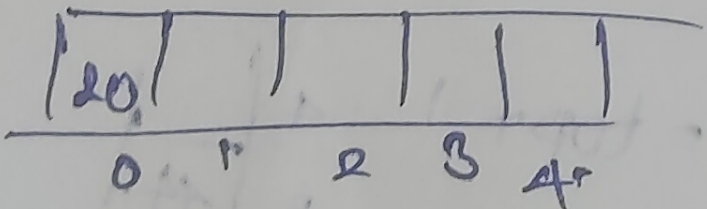
$$= 1$$

$$[Stack] = 22$$

Q. Perform the following operations in circular queue having

size of 5: Enqueue(20), Enqueue(10), En(30), Dq(),
En(80), Dq(), Enq(90), Enq(100), Dq(), Dq(), Enq(60), Enq(90),
En(80), Dq(), Enq(90), Enq(100), Dq(), Dq(), Enq(60), Enq(90).

i) Enq



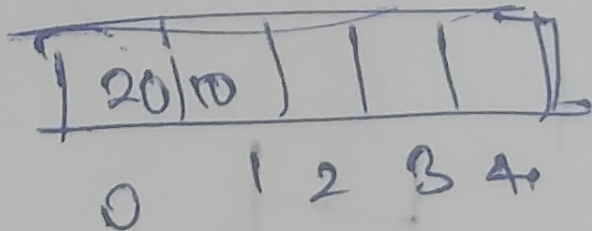
$$Front = 0$$

$$Rear = rear + 1$$

$$= -1 + 1$$

$$= 0$$

ii) En(20)



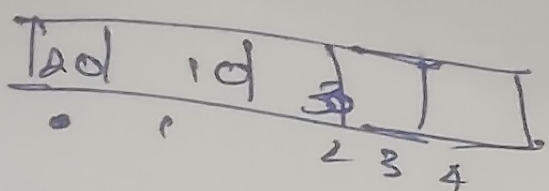
$$Front = 0$$

$$Rear = rear + 1$$

$$= 0 + 1$$

$$= 1$$

iii) En (30).



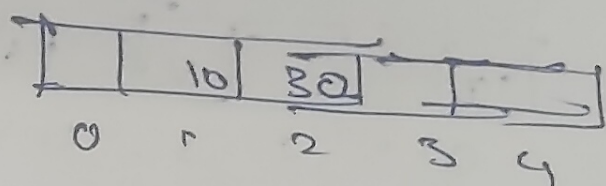
$$\text{Front} = 0$$

$$\text{Rear} = \text{Rear} + 1$$

$$= 1 + 1$$

$$= 2$$

iv) De (2).

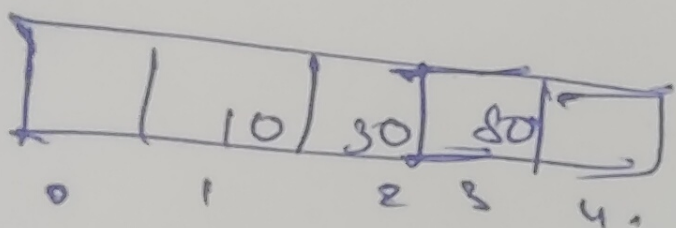


$$\text{Fr} = \text{Fr} + 1$$

$$= -1 + 1$$

$$= 0$$

v) En (80)



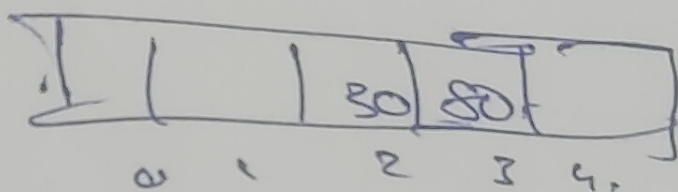
$$\text{Rear} = 0$$

$$= \text{Rear} - 1$$

$$= 2 - 1$$

$$= 1$$

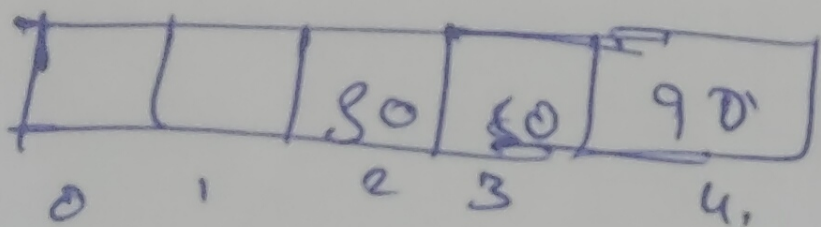
vi) De (2)



$$\text{Fr} = 1$$

$$\text{Rear} = 0$$

vii) En (100)



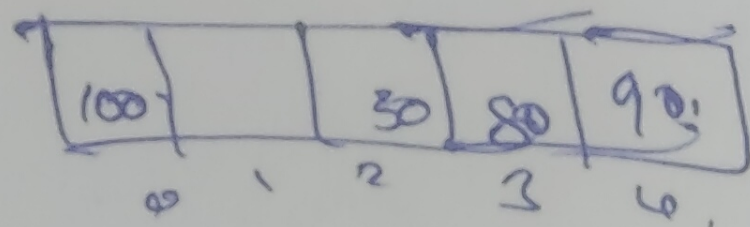
$$\text{Front} = 1$$

$$\text{Rear} = \text{Rear} + 1$$

$$= 3 + 1$$

$$= 4$$

vii) De (100)



$$\text{Front} = 1$$

$$\text{Rear} = -1 + 1$$

$$= 0$$

Convert the following infix expression into postfix expression

$$B - (C / D + (E \times F * G) / H) + J$$

Scan Symbol

Stack

Postfix

B

B

B

-

B -

B -

(

B (

B (

C

B (C

B C

/

B (C /

B C /

D

B (C / D

B C D

+

B (C / D +

B C D /

(

B (C / D + (

B C D / (

E

B (C / D + (E

B C D / E

%

B (C / D + (E %

B C D E %

F

B (C / D + (E % F

B C D E F %

*

B (C / D + (E % F *

B C D E F % *

G

B (C / D + (E % F * G

B C D E F % * G

)

B (C / D + (E % F * G)

B C D E F % * G +

)

B (C / D + (E % F * G) +

B C D E F % * G +

/

B (C / D + (E % F * G) + /

B C D E F % * G + /

H.	<u> </u>	BCD/EF % 01 + 11 /
*	<u> </u> *	BCD/EF % 01 * + 11 /
J	<u> </u> *.	BCD/EF % 01 * + 11 /
End	<u> </u>	BCD/EF % 01 * + 11 /

BCD/EF % 01 * + 11 /

① 9384 / * +

Symbol

9
3
8
/
*
+

Stack

9
93
938
9384
932
96
15

Answer: 15

② 62 + 63 / *

Symbol

6
2
+
6
3
/
*

Stack

6
62
8
86
863
82
16

Ans: 16

dy

10	1	10	90
0.1	2	3	4

$$F_{\text{m}} = 1 + 1$$

$$= 2$$

$$R_{\text{m}} = 0$$

Not
confused.